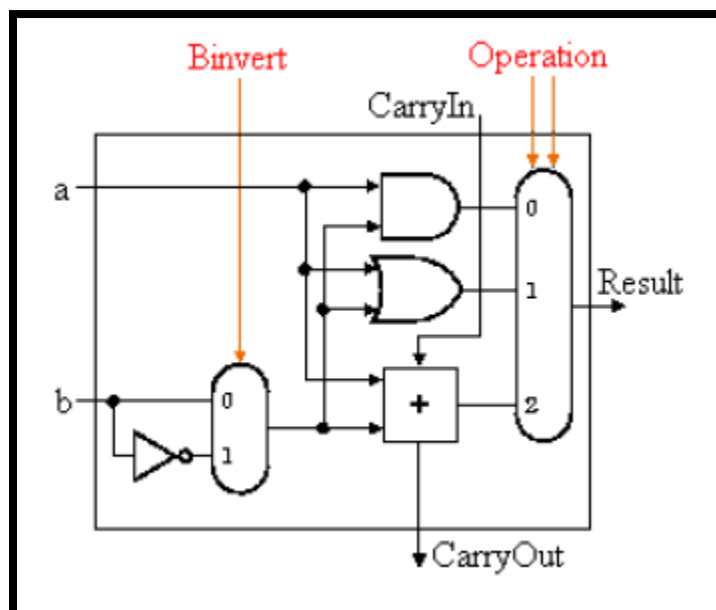


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Q1) Complete the Following Table with Control Lines to Achieve this 4 Operation:



Control Lines	Operation
	<i>SUB</i>
	<i>ADD</i>
	<i>AND</i>
	<i>OR</i>

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**Q2) Express the Following HexaDecimal to Decimal Using IEEE 754
Floating Point Representation:**

0x40F00000	0xE4F01100

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**Q3) Based on The Final Division Algorithm, Complete
0111(two)/0011(two) by Filling in The Following Table:**

Iteration	Divisor (D)	Reminder (R)	Remark

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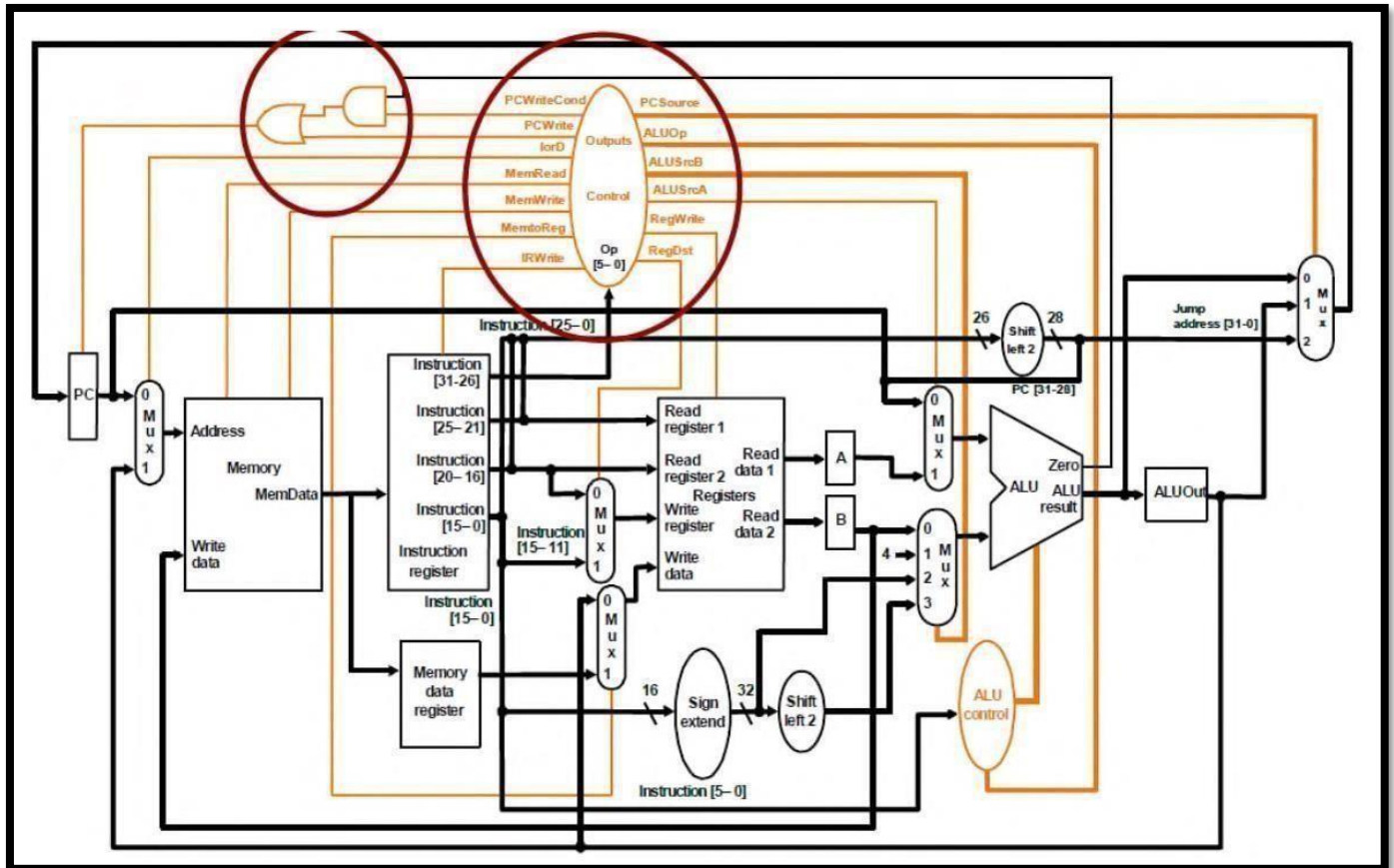
Q4) Show how the Instructions in the sequence given below will proceed through the pipeline using forwarding method :

**** Hint : we will predict that the beq instruction isn't taken.**

	1	2	3	4	5	6	7	8	9	10	11	12	13	14
														
beq \$1,\$2,X														
lw \$10,0(\$11)														
sub \$14,\$10,\$10														
X: add \$4,\$1,\$2														
lw \$1,0(\$4)														
sub \$1,\$1,\$1														
Add \$1,\$1,\$1														

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(50 Marks)

Q5) Show the Signal in The Cycle Number 1 in Multicycle Datapath:



<i>Signal</i>	<i>Value</i>
<i>PCWrite</i>	
<i>lorD</i>	
<i>MemRead</i>	
<i>MemWrite</i>	
<i>IRWrite</i>	
<i>PCSource</i>	
<i>ALUOp</i>	
<i>ALUSrcB</i>	
<i>ALUSrcA</i>	
<i>RegWrite</i>	

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Q6) Assume MIPS Big Endian for the Byte Order Convention in this Problem, The Following Figure Shows the content trunk of the memory with the corresponding addresses.

**** Hint : Assume the value represented in the memory are signed and in 2's complement form.**

**** The Value Of Register \$s0 = 0 , \$s1 = 4 and \$s2 = 9**

- 1) What's the content of registers \$t0,\$t1,\$s1,\$s2 after executing the program ?**
- 2) Show The content of the same trunk of memory after executing the program ?**

lw \$t0,0(\$s1)

lw \$t1,-9(\$s2)

mult \$t1,\$t0

mflo \$s1

sw \$s1,0(\$s0)

mfhi \$s2

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Before :

Address	Content
0x7	0x01
0x6	0x00
0x5	0x00
0x4	0x00
0x3	0x0d
0x2	0x00
0x1	0x00
0x0	0x00

After :

Address	Content
0x7	
0x6	
0x5	
0x4	
0x3	
0x2	
0x1	
0x0	

\$t0	
\$t1	
\$s1	
\$s2	

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Q7) Compute the following information about ram executing in two Computer (Computer 1 & Computer 2) and find performance ISA.

**** Hint :** السؤال صيغته مش 100% صحیحة لكن المهم الفائدة انك تتاكد انك ترقى السرعة بين التايين :

	Computer 1	Computer 2
IC	$2.0 * 10^9$	$3.0 * 10^9$
CR	1 GHz	2 GHz
CPI	2	4

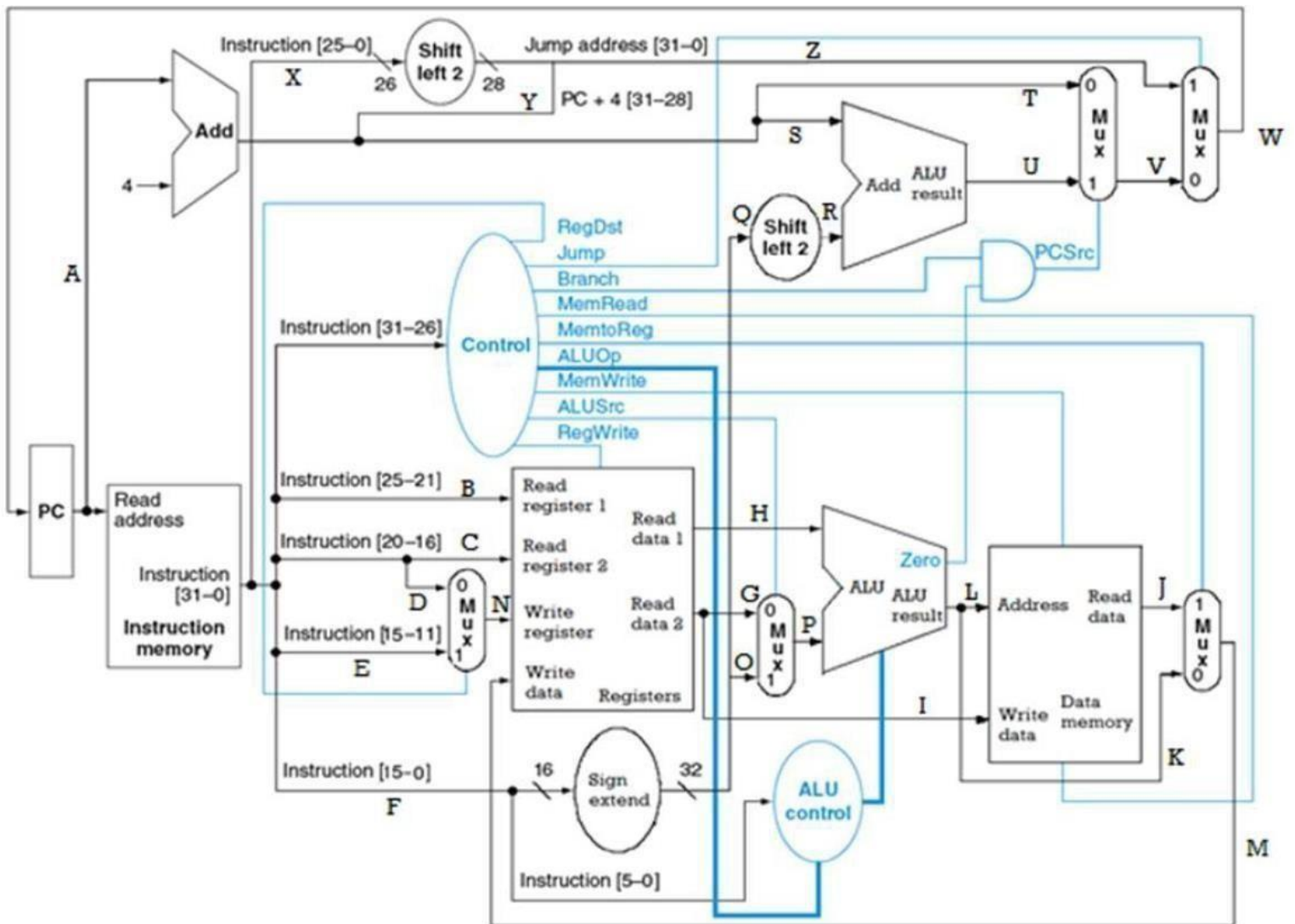
- **Find Which Computer Is Faster , Then Calculate How Much Is Faster Than Another Computer ?**

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Computer Architecture and Organization Final Exam (50 Marks)

Q8) Consider the following MIPS single cycle processor, and the snapshot of the instruction memory, data memory and the register file.

Execute the instructions and fill the following table (Buses and Control Signals), assuming that the PC register contains 0x00400004.



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Instruction Memory		
Address	Contents	Instruction
0x00400000	0x0253A825	OR \$S5, \$S2, \$S3
0x00400004	0x02309020	ADD \$S2, \$S1, \$S0
0x00400008	0x12530001	Beq \$S2, \$S3, 1
0x0040000c	0x8E93FFFE	LW \$S3, -2(\$S4)
0x00400010	0x08100001	J 0x100001

Register File	
Register#	Data
16 (\$s0)	0x0000000A
17 (\$s1)	0x00000001
18 (\$s2)	0x02E40160
19 (\$s3)	0x00000011
20 (\$s4)	0x00020006
21 (\$s5)	0x00012304

Data Memory	
Address	Data
0x00020002	0x13
0x00020003	0x2E
0x00020004	0xA4
0x00020005	0xFF
0x00020006	0x08
0x00020007	0x2C

Instr. Buses					
A					
B					
C					
D					
E					
F					
G					
H					
I					
J					
K					
L					
M					
N					
O					
P					
Q					
R					
S					
T					
U					
V					
W					
X					
Y					
Z					

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(50 Marks)

Control Sig.					
RegDst					
RegWrite					
ALUSrc					
MemRead					
MemWrite					
MemtoReg					
Jump					
Branch					
ALUOpr1					
ALUOpr0					
PCSrc					

Registers					
16 (\$s0)					
17 (\$s1)					
18 (\$s2)					
19 (\$s3)					

20 (\$s4)					
21 (\$s5)					

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Q9) Consider Two Different Implementations, M1 and M2, of the same Instruction set. There are three classes of Instruction (A,B and C) in the instruction set. M1 has a Clock Rate of 80 MHz and M2 has a clock rate of 100 MHz, The Average Number of cycles for each instruction class and their frequencies (for a typical program) are as follows:

Instruction Class	M1 – Cycles/Instruction Class	M2 – Cycles/Instruction Class	Frequency
A	1	2	60%
B	2	3	30%
C	4	4	10%

A) Calculate the average CPI for each machine, M1 and M2.

B) Calculate the average MIPS rate for each machine, M1 and M2.

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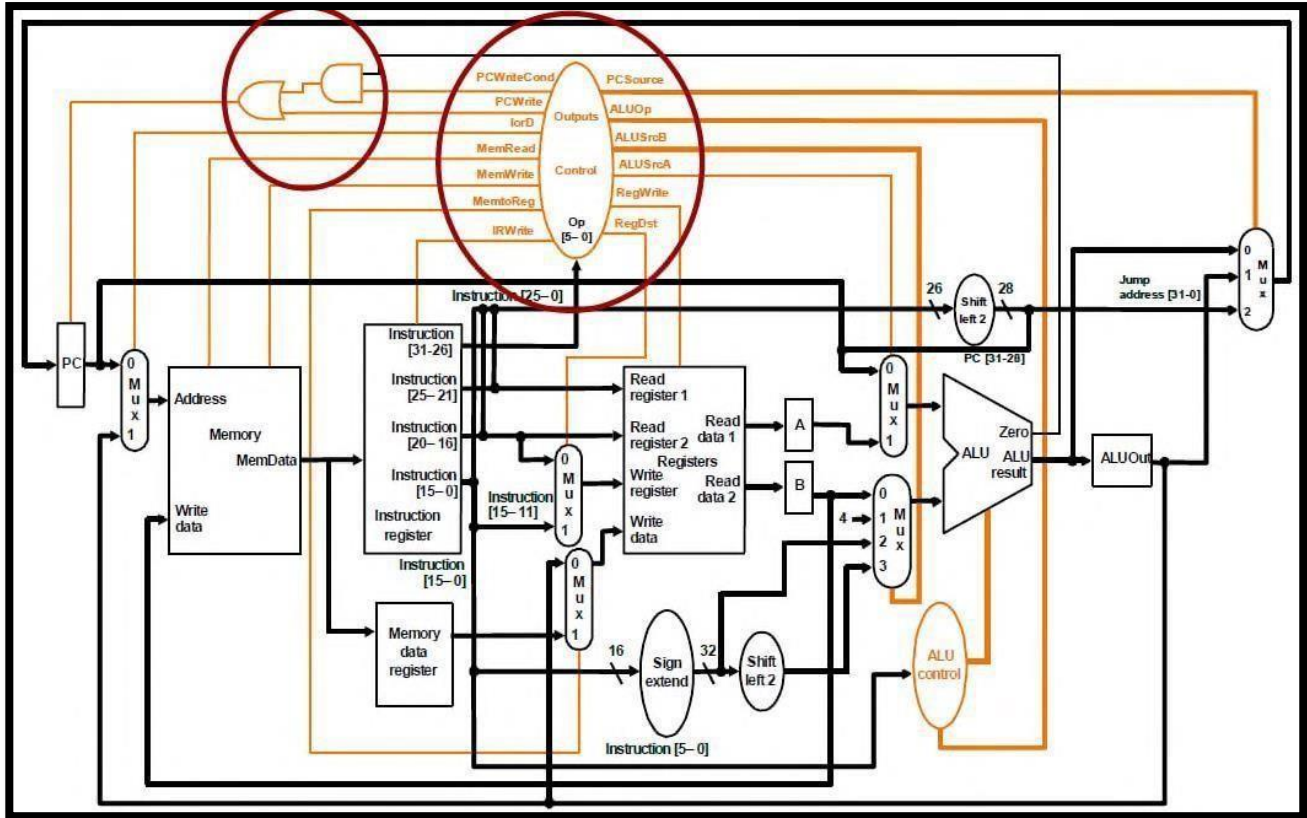
Q10) Find The Value of \$s2 Register After Execution the following Instruction ..

**** Hence the Value of \$s0 = 0x00000010 and \$s1 = 0x00000020.**

Instruction:	Solution:
and \$s2,\$s1,\$s0	
or \$s2,\$s1,\$s0	
slt \$s2,\$s0,\$s1	
sll \$s2,\$s0,2	
addi \$s2,\$s0,30	

Computer Architecture and Organization Final Exam (50 Marks)

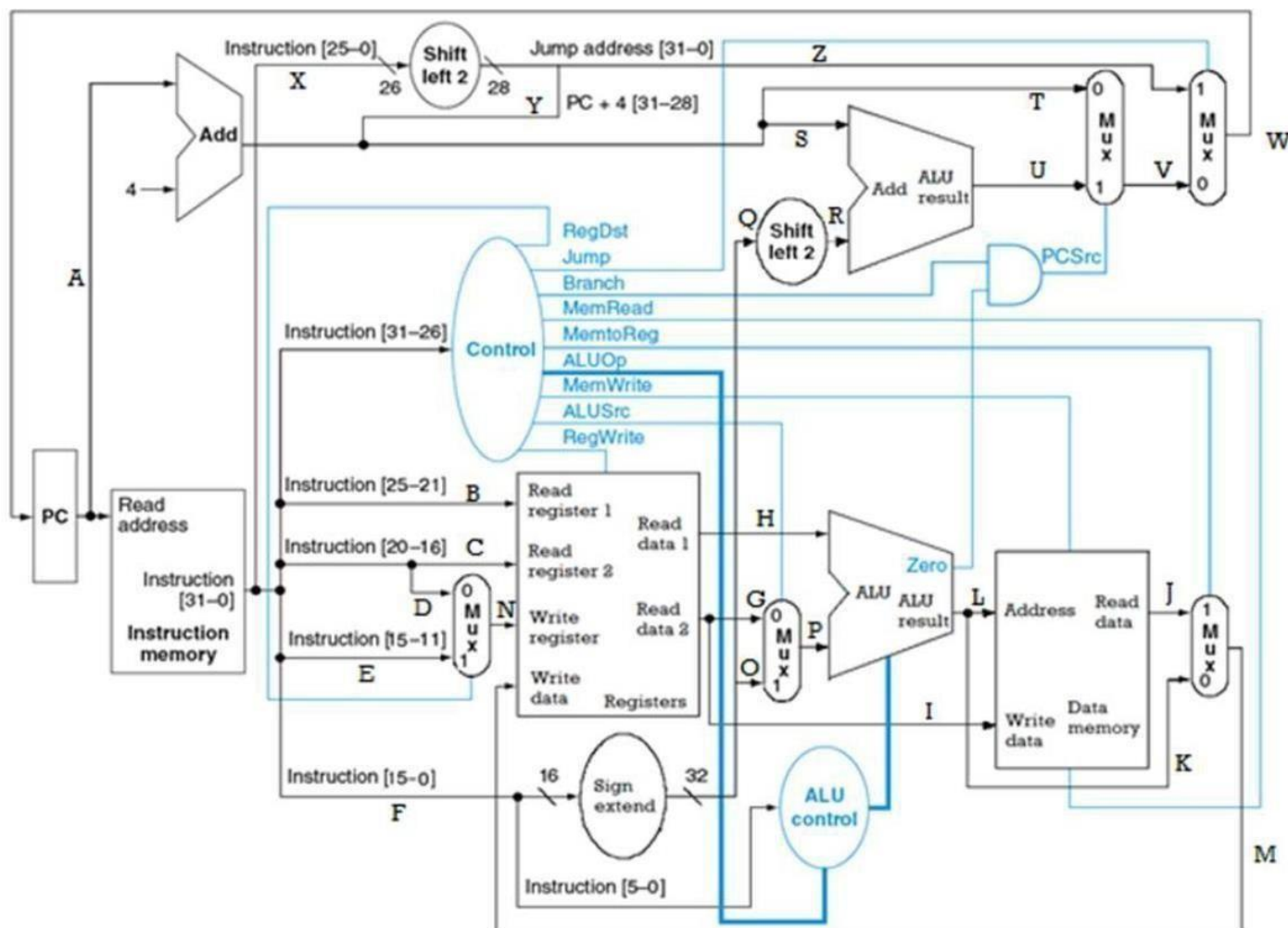
Q11) In the Multicycle Datapath Draw or Show the lines in (Beq) Instruction for 3 Cycle, Add Control line If you need to.



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Computer Architecture and Organization Final Exam (50 Marks)

Q12) Consider the following MIPS single cycle processor, Find The Control Signal For SW (Store) and ADDI (Add Immediate) Instructions.



Instr.	RegDst	RegWrite	ALUSrc	MemRead	MemWrite	MemtoReg	Jump	Branch	ALUOp1	ALUOp0	PCSrc
SW ...											
addi ...											

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